

From Automation to Decision-Making: Reflections on the Development of TechCorp Decision Game

Introduction

Throughout this module, I explored a series of themes concerning technology, artificial intelligence, complexity, prediction, and human decision-making. These ideas were examined in several essays discussing communication, digital systems, artificial intelligence, predictive governance, and the automation of creative processes. My final project, TechCorp Decision Game, provided an opportunity to translate many of these theoretical concepts into a practical software application.

TechCorp Decision Game is a turn-based business simulation written in Java. The player manages a technology company, hires employees, develops projects, responds to changing circumstances, and competes against an AI-controlled rival company. While the project was primarily designed to demonstrate object-oriented programming concepts, the development process also encouraged me to reflect on broader questions concerning systems, decision-making, and the relationship between humans and technology.

This essay discusses how the project connects to several themes explored throughout the module and reflects on what I learned from both the technical and conceptual aspects of the course.

Complexity, Simplicity, and Software Design

One recurring theme throughout the module was the relationship between complexity and understanding. Complex systems often appear simple from the outside because their internal structures are hidden from users.

This idea strongly influenced the design of my project. From the player's perspective, the game is simple. The player chooses actions from a menu, hires employees, and develops projects. However, behind these choices exists a collection of interacting classes, objects, methods, and game rules.

To manage this complexity, I applied object-oriented programming principles. Classes such as Company, Employee, Project, and GameEngine separate responsibilities into manageable units. Encapsulation allows each class to manage its own data, while inheritance and polymorphism allow different employee types to share common behavior while maintaining unique characteristics.

This design process demonstrated an important lesson from the module: simplicity is not the absence of complexity. Instead, effective design organizes complexity in a way that allows users to interact with a system without needing to understand every internal detail.

The Company as a System

Another major theme explored during the course was the idea of systems and environments that influence behavior.

In TechCorp Decision Game, the company functions as a system rather than a collection of isolated components. Employees generate productivity, projects create revenue, reputation influences company value, and random events affect performance. Every decision influences multiple variables simultaneously.

The game therefore models a simplified version of organizational decision-making. Players may believe they are making independent choices, but those choices occur within constraints created by available resources, employee skills, project requirements, and economic conditions.

This reflects a broader understanding of how modern technological systems operate. Decision-making is rarely completely free. Instead, choices are made within structures that shape available possibilities.

Artificial Intelligence and Competition

Artificial intelligence was one of the central topics of this module. In my project, an AI-controlled company competes against the player. The AI makes decisions automatically by hiring employees and progressing through projects.

Although the AI in the game is intentionally simple, implementing it encouraged me to think about the role of automated decision-making. The AI follows rules and patterns rather than genuine understanding. However, from the player's perspective, it still creates competition and uncertainty.

This mirrors a larger discussion surrounding artificial intelligence. Many modern AI systems generate useful outputs without possessing human consciousness or intention. Their effectiveness comes from processing patterns rather than understanding reality directly.

Developing the AI competitor helped me appreciate how relatively simple algorithms can produce meaningful interactions within a larger system.

Prediction and Business Management

The module also explored prediction and the increasing use of data to anticipate future outcomes.

Several aspects of the game incorporate predictive thinking. When hiring employees, players invest resources based on expected future benefits. Choosing whether to spend money now or save resources for later requires forecasting future outcomes. Similarly, project development involves balancing short-term costs against long-term rewards.

Although the game does not use advanced predictive algorithms, it simulates the logic of prediction that exists within many real-world organizations. Business decisions are often based on probabilities and expectations rather than certainty.

This aspect of the project helped me understand how software systems frequently support decision-making by organizing information about possible futures rather than providing guaranteed answers.

Automation and Creativity

One of the most interesting themes discussed during the course concerned the automation of creation.

As a programming project, TechCorp Decision Game itself emerged from a process of learning patterns. During the module, I learned programming concepts, design principles, testing strategies, and software architecture techniques. These patterns were gradually combined and adapted to create a functioning application.

This experience reinforced the idea that software development is both technical and creative. Writing code involves solving problems, designing systems, and creating new structures from existing knowledge. The project therefore demonstrated that creativity in programming often emerges through the transformation and recombination of learned patterns.

The development process also highlighted how modern tools increasingly assist programmers. Development environments, documentation, testing frameworks, and AI-assisted tools all contribute to software creation. Rather than replacing creativity, these technologies often extend it.

Technical Learning Outcomes

From a technical perspective, the project allowed me to apply many core concepts from the Java Programming module.

These included:

- * Object-Oriented Programming
- * Encapsulation
- * Inheritance
- * Polymorphism
- * Interfaces
- * Enumerations
- * Exception Handling
- * Maven Project Management
- * Unit Testing with JUnit

Software Design and Modularity

The project structure was organized into multiple packages, including domain, engine, events, user interface, and testing components. This modular design improved maintainability and demonstrated the importance of separating responsibilities within software systems.

Testing also played an important role in development. Creating JUnit tests helped verify that key components behaved as expected and improved confidence in the overall reliability of the application.

Conclusion

The development of TechCorp Decision Game provided more than an opportunity to practice Java programming. It also served as a practical reflection on many of the broader themes explored throughout this module.

The project demonstrated how complex systems can be organized through abstraction, how decision-making occurs within structured environments, how automated agents influence outcomes, and how software creation itself depends upon learned patterns and accumulated knowledge.

Most importantly, the project showed that programming is not only about writing code. It is also about designing systems, modeling relationships, and creating tools that help people understand and interact with complexity.

Through this project, I gained both technical skills and a deeper appreciation for the wider social and philosophical questions surrounding technology and artificial intelligence. The combination of theory and practical development has significantly strengthened my understanding of software engineering and its role within

contemporary society.